

Remote Distribution — `lava-api-go to thinker.local`

The Lava Go API service can run on a dedicated LAN host instead of the operator's workstation. This document describes the SSH-based distribute flow used by `scripts/distribute-api-remote.sh`.

Why a remote host

- **Frees the operator's workstation** from running Postgres + the Go API + its emulator + IDE + browsers + everything else simultaneously.
- **One LAN-resident API** — Android testers running 1.2.6+ discover the API via mDNS regardless of which workstation is currently active.
- **Mirrors the Catalogizer + HelixAgent pattern** for vasic-digital services running on `thinker.local`.

Prerequisites (one-time setup)

1. **Passwordless SSH** from the operator's workstation to the remote host:

```
# On the operator's workstation:
ssh-copy-id milosvasic@thinker.local
ssh milosvasic@thinker.local 'true' && echo "OK"
```

(If it prints OK without prompting for a password, you're done.)

2. **Rootless Podman 4.x or later** on the remote host:

```
ssh milosvasic@thinker.local 'podman --version'
# Expected: podman version 4.x.y or 5.x.y
```

3. **.env keys** in the operator's workstation `.env` (gitignored — see `.env.example`):

```
LAVA_API_GO_REMOTE_HOST=thinker.local  
LAVA_REMOTE_HOST_USER=milosvasic
```

These default to the same values as the example, so most operators don't need to override them.

Distribute

```
# Build (if needed) + ship + boot the API on the configured remote  
# host:  
./scripts/distribute-api-remote.sh  
  
# Or target a different host:  
./scripts/distribute-api-remote.sh another-host.local  
  
# Or use a different SSH user:  
./scripts/distribute-api-remote.sh thinker.local --user other-user
```

The script:

1. Verifies SSH connectivity + Podman presence on the remote.
2. Resolves the lava-api-go version from lava-api-go/internal/version/version.go.
3. Runs `./build_and_release.sh` if the image tarball is missing.
4. Copies the OCI image tarball, `deployment/thinker/thinker.local.env`, `deployment/thinker/thinker-up.sh`, and TLS material (`server.crt` + `server.key`) to the remote.
5. Runs `thinker-up.sh` on the remote, which:
 - Creates the lava-thinker Podman network if missing.
 - Loads the image tarball + tags as `localhost/lava-api-go:thinker`.
 - Recreates lava-postgres-thinker (Postgres 16-alpine).
 - Recreates lava-api-go-thinker (the API binary built from lava-api-go/).
 - Waits for `/health` on the remote to respond.
6. Verifies `https://$REMOTE_HOST:8443/health` from the operator's workstation as a final end-to-end probe.

Verify (from the workstation)

```
curl -fsSk https://thinker.local:8443/health  
# Expected: {"status":"alive"}  
  
curl -fsSk https://thinker.local:8443/ready  
# Expected: {"status":"ready"}
```

Or remote-side:

```
ssh milosvasic@thinker.local 'podman ps --filter name=lava-'  
ssh milosvasic@thinker.local 'podman logs --tail 30 lava-api-go-  
thinker'
```

Tear-down

```
./scripts/distribute-api-remote.sh --tear-down thinker.local
```

Removes the lava-api-go-thinker + lava-postgres-thinker containers, the localhost/lava-api-go:thinker + localhost/lava-api-go:dev images, and the lava-thinker Podman network. Idempotent.

Re-deploy on every change

Per §6.P, the distribute scripts refuse to operate without a versionCode bump + matching CHANGELOG.md entry. So re-distributing after changes is:

```
# 1. Bump lava-api-go version in lava-api-go/internal/version/  
version.go  
# 2. Add CHANGELOG.md entry + per-version snapshot at  
# .lava-ci-evidence/distribute-changelog/container-registry/  
<v>-<code>.md  
# 3. Rebuild + redistribute:  
./build_and_release.sh  
./scripts/distribute-api-remote.sh
```

Constitutional bindings

- **§6.J** Anti-Bluff — scripts/distribute-api-remote.sh and deployment/thinker/thinker-up.sh use set -euo pipefail; failures propagate.
- **§6.B** "Container Up" is not healthy — the script verifies /health end-to-end, not just podman ps.
- **§6.K** Builds-Inside-Containers — image is produced by build_and_release.sh which routes through the container build path.
- **§6.M** Host-Stability — rootless Podman; no host power-management commands.
- **§6.H** Credential Security — credentials come from .env (gitignored); never logged.

- **§6.P** Distribution Versioning + Changelog Mandate — every distribute requires a versionCode bump + CHANGELOG entry + per-version snapshot.

Reference projects

- `../HelixAgent/scripts/deploy-full-remote-pass.sh` — fuller multi-service pattern with profile support
- `../Catalogizer/deployment/thinker.local.env` + `../Catalogizer/deployment/thinker-up.sh` — the closest analog; this Lava implementation follows the same shape